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4

Trueness

4.1 Trueness and True Value

The Bureau International des Poids et Mesures (BIPM) oversees developing and promoting metrology, which is "the set of techniques and know-how that allow measurements to be made and to have sufficient confidence in their results." It is therefore one of the keys to assessing trueness. Analytical chemists often stay away from metrology, which represents a considerable challenge in terms of science, economics, and societal demands. This can be explained by the fact that analyst concerns were only recently addressed by metrologists. Several tools developed by the BIPM to assess trueness in analytical sciences are presented in Section 4.2 but for many decades, the question of measuring the "amount-of-matter" was neglected by the BIPM. Recently, a strong effort has been made, and many solutions are now proposed.

While there is a continuum of parameters for expressing precision, ranging from repeatability to reproducibility through intermediate precision standard deviations, the estimation of trueness is much simpler, in theory. Whatever the parameter used, they all represent the same idea:

> "the closeness of agreement between the average of an infinite number of replicate measures of a quantity values and a reference value."

Although the International Vocabulary of Metrology (VIM) is very explicit, the term trueness is sometimes confounded with accuracy. Accuracy is the "closeness of agreement between a measured value and a true value of a measurand." According to the VIM philosophy, accuracy is a concept and is not quantifiable, while trueness is more practical and measurable. This difference is better explained in Section 5.1.1 about the vocabulary used in method validation.

The challenge in assessing trueness is to obtain the adequate reference value or true value, which is denoted X throughout this book. For some analysts, this is such a tricky issue that they may think it is impossible, for instance, when the analyte is defined by the method itself, as it is in food microbiology.

Quantification, Validation and Uncertainty in Analytical Sciences: An Analyst's Companion,
First Edition. Max Feinberg and Serge Rudaz.

4.1.1 Bias and Recovery Yield

Trueness raises the question of the definition of the analyte. It was partly discussed in Chapter 1 about quantification. To introduce trueness parameters, let us keep the notations already used, namely:

– X the target (or reference or assigned) value of a sample.
– $\overline{Z}$ the average of replicates Z_i obtained for this sample.

The three parameters listed here are common to express trueness, They do not really measure the trueness but rather the lack trueness. This is why they are called as bias, except the recovery yield
Bias

$$\delta = \overline{Z} - X \tag{4.1}$$

Relative bias

$$\delta\% = \left(\frac{\overline{Z} - X}{X}\right) \times 100 \tag{4.2}$$

Recovery yield

$$RY\% = \frac{\overline{Z}}{X} \times 100 = 100 - \delta\% \tag{4.3}$$

They are roughly equivalent. If the value of the bias or relative bias is negative, it means that the method underestimates the target concentration, but if it is positive, the method overestimates. For the recovery rate, a comparison must be made with respect to 100%. For analysts who are pleased to utilize percentages, the recovery rate is the most meaningful. The relative bias is the complement of the recovery yield and can be interpreted as a correction factor that measures the proportion of what was measured compared to what was expected to be measured.

Most regulatory documents also propose percentages as acceptance criteria. Similarly, precision can be expressed as a percentage by estimating relative standard deviations, as explained in Section 3.2.1; they define trueness by means of the recovery yield. An example is given in Section 7.5.1.

Below, the worksheet excerpt illustrates these parameters and data in cells `B5` and `B6`, showing that in the worksheet it is not useful to multiply results by 100 to obtain a percentage. The editing format "Percent" directly gives the expected value and adds the "%" symbol. This mode of expression is preferable because the parameter may be used later, for instance, to correct a measurement value. It is easier as it is not necessary to multiply or divide the data by 100.

	A	B	C
1	**Trueness**		
2	Reference value	14	
3	Measurement value	13.6	
4	Absolute bias	–0.4	=B3 - B2
5	Relative bias	–2.86%	=(B3 - B2)/B2
6	Recovery yield	97.14%	=B3/B2

For some methods it is conventional to transform measurements into decimal logarithms, for instance, the microbiological counting methods. In this case, absolute bias is equivalent to the recovery yield since the logarithm of a ratio is equal to the difference of the logarithms.

Recovery yield in log

$$\log\left(\frac{\overline{Z}}{X}\right) = \log(\overline{Z}) - \log(X) \tag{4.4}$$

4.1.2 Evolution of the Concept of True Value

In the 1980s, BIPM decided to address the question of measurement error and proceeded from what is called the traditional "error approach" to the new uncertainty approach, and to reconsider some extensively used concepts.

The rationale behind the error approach was to determine an estimate of the true value that was as close as possible to the theoretical true value. This was based on the previous definition of accuracy. In this context, the true value was considered unique and, in practice, unknowable. It was assumed that the deviation from the true value was a combination of different sources of error. To differentiate the sources of error, the concepts of fixed error and random systematic error were introduced as always distinguishable and to be treated differently. For example, systematic error may be due to the use of poorly graduated glassware, while random error is caused by a drifting measuring device that does not always give the same value.

In fact, it is often recognized a systematic error can be claimed as fixed when its variations in amplitude are small when compared to the measurement value, but it becomes random when they are large enough to introduce measurable fluctuations. It is traditional to say that the categories of systematic error vary according to the principles of the method. Controlling them is part of the analyst's expertise: the development of the method must make it possible to highlight and correct them. Thus, a bias can have various origins, such as:

- Evolution of the sample over time.
- Degradation of calibration solutions.
- Modifications of equipment settings.
- Miscalculations.
- Human errors.

Figure 4.1 illustrates a source of systematic error frequently encountered in chromatography. It could be related to the integration method used to calculate the areas of poorly resolved peaks. In the graph, two peaks are represented, centered on retention times of 2.0 and 3.1 min, respectively, with a drift of the baseline. The peak of interest is the second, smaller peak.

The method used is the drop-line algorithm, symbolized by the dashed lines. It consists in locating the beginning and end of each peak, considering the drift, and then assigning to each one a part of the common triangular overlapping area, by drawing the perpendicular to the point of separation of the peaks. In the graph on the left, since the surfaces of the two peaks are quite close to each other, the error remains negligible. However, it becomes significant when the area of the peak of interest decreases, as in the graph on the right.

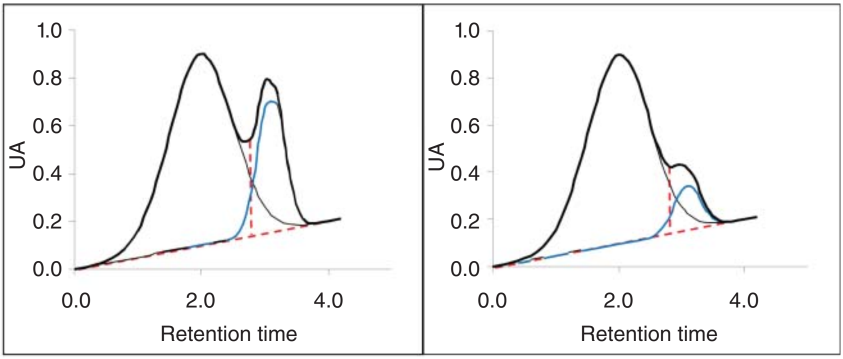

Figure 4.1 Example of systematic error generated by the integration mode of poorly resolved chromatographic peaks.

Furthermore, if the major peak is very asymmetric, the situation becomes worse. The uncertainty approach is to recognize that, owing to the inherently incomplete amount of detail in the definition of a quantity:

> "there is not a single true value but rather a set of true values consistent with the definition."

However, the complete set of values is, in principle and in practice, unknowable. However, it is possible to estimate an interval containing a given proportion of these true values. The change in the paradigm of the single, unique true value to the bounds of an interval is an especially important fact for experimenters. More details on the consequences are described in Sections 4.3.2 and 6.3.

4.1.3 Specificity and Sources of Bias

In the context of analytical sciences, specificity is "the capability of a measuring system or operating procedure to measure the concentration of a given analyte." The main problem when addressing specificity is the potential presence of interfering compounds that alter the analyte's detection integrity. Analytical literature is extensively dedicated to the topic of the various chemical, physical, or biological mechanisms that are the source of interference and specific to each method. For example, in atomic absorption spectrophotometry, spectral interferences can be due to the lack of resolution of monochromators, or the broadening of the absorption bands caused by the earth's magnetic field and matrix interferences to various elements absorbing at the same wavelength as the searched analyte.

Figure 4.2 is a classic representation of bias, which is made possible when considering how the inverse-predicted concentration is computed (Section 2.2). First, let us define the first bisecting line (dotted line) as the *trueness line* where measurements are not biased, i.e. the inverse-predicted concentration is exactly equal to the known reference concentration. Then consider a collection of materials whose reference values are known. Once they are